

Project Report: SpendSmart Expense Tracker

Course: ICS 2300 Mobile Applications Design and Development

Group: Group 2

Technical Lead: Shemiramoth Mugo

1. Project Overview & Justification

The **SpendSmart** application is a modern personal finance tool designed to solve the real-world problem of "invisible spending" among young adults. By providing a categorized logging system and visual progress bars, the app encourages financial literacy and prevents over-expenditure. The official project repository is maintained at: <https://github.com/Shemira01/MADD-Project-2.git>.

2. Global App Architecture

The application is built using **Java** and follows a robust development structure:

- **Structure:** Single Activity Architecture where MainActivity serves as the primary host.
 - **Navigation:** A BottomNavigationView using a dedicated Menu Resource for seamless transitions.
 - **Fragment Management:** Dynamic swapping using getSupportFragmentManager and replace() transactions.
 - **Design Language:** Material Design 3 (M3) utilizing TextInputLayout.OutlinedBox and custom color mapping (colorPrimary to primary) for a modern, professional interface.
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3. Page-by-Page Breakdown

Page 1: Dashboard (DashboardFragment)

- **UI Components:** MaterialCardView for Balance display and a ProgressBar for visual budget status.
- **Content:** A RecyclerView to display recent transactions.

Page 2: Add Expense (AddExpenseFragment)

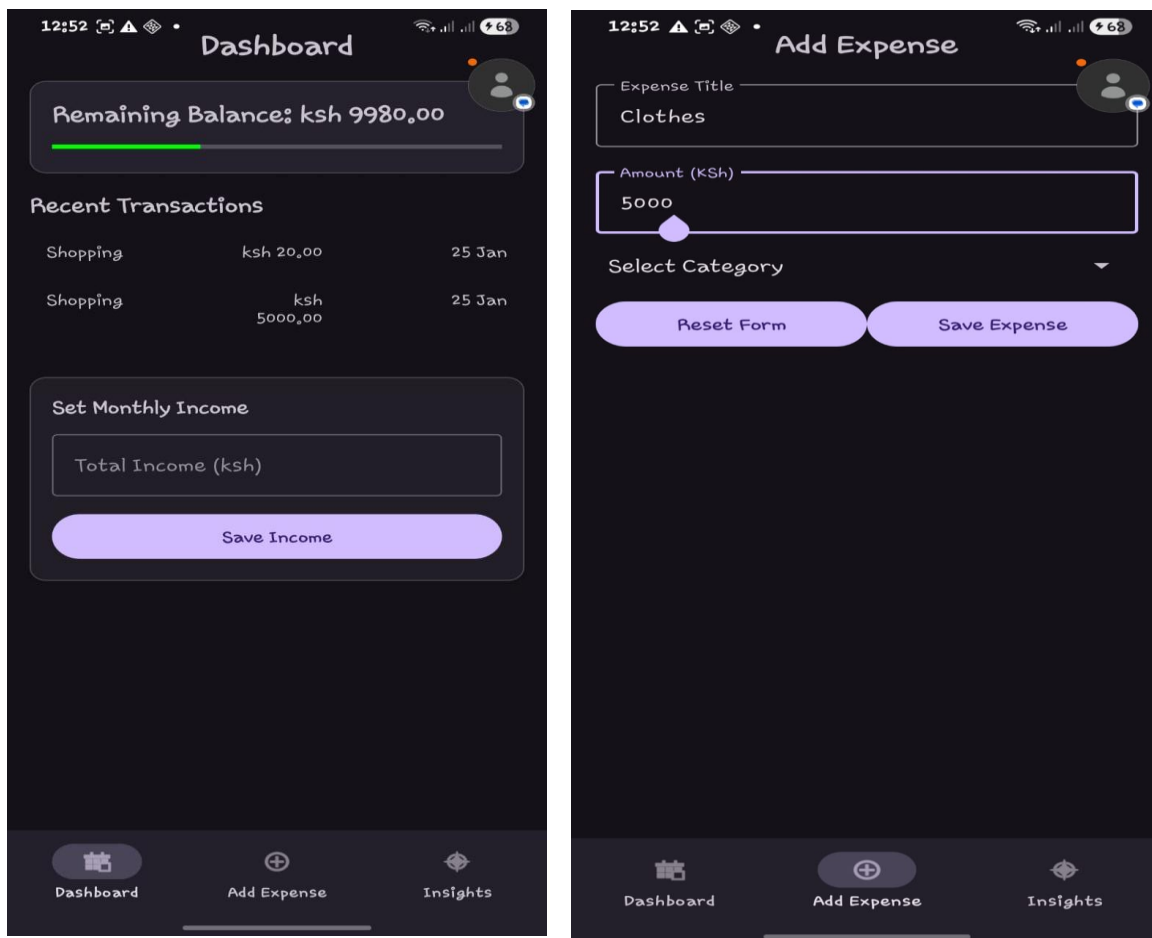
- **UI Components:** TextInputLayout (Outlined style), TextInputEditText, and a Spinner for categories.
- **Key Feature:** resetForm() – This programmatically clears all inputs and returns cursor focus to the amount field to facilitate high-speed data entry.

Page 3: Insights (InsightsFragment)

- ❑ **UI Components:** A statistical breakdown of spending categories using horizontal progress indicators to show budget consumption.

4. Implementation & Physical Device Testing

The application was successfully deployed and tested on a physical mobile device to verify Material 3 responsiveness and form logic.



5. Team Roles & Task Allocation

6. Conclusion

The **SpendSmart** project successfully transitioned from a conceptual blueprint to a functional mobile application. By utilizing Java and the Single Activity Architecture, the team delivered a tool that effectively visualizes personal finances. Deployment on a physical device confirmed that the UI remains intuitive and the data validation logic is stable for real-world use.

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